Course Outline for CSE- 345

Part A

1. Course Code: CSE-345

2. Course Title: Artificial Intelligence

3. Course Type: Core Course

4. Level/ Term: Level: 3 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Mohammed Moshiul Hoque, Professor, Dept. of CSE, CUET
Ashim Dey, Assistant Professor, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 3

9. Contact Hours: 3 lectures of 50 minutes per week

10. Total Marks: 300

11. Rationale of the Course:

This course will cover the knowledge representation, problem solving, and learning methods of Artificial Intelligence. The primary focus of this course is to solve complex problems based on the intelligent behavior of humans and other animals using state-space search techniques, rules, expert systems, frames, semantic nets, and propositional and first-order predicate logic. At the end of the course students will be able to apply the intelligent agent paradigm as an underlying model for exploring AI problem domains, tools, and techniques. This is a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalog description of the course is

Course Content:

Introduction to old and new AI techniques; Intelligent agents; Search techniques in AI; Constraint satisfaction problems; Game playing; Knowledge representation and reasoning; Propositional and first order logic, inference in first order logic; Planning;

Uncertainty; Probabilistic reasoning; Learning in symbolic and non-symbolic representation; Expert systems and knowledge engineering; Natural language processing; Computer vision; Robotics;

12. Course Objectives:

- Explain the basic concepts of Artificial Intelligence including Search, Game Playing, Knowledge Based System (including Uncertainty), Planning and Machine Learning.
- Assess the applicability, strengths, and weaknesses of the basic knowledge representation, problem solving, and learning methods in solving particular engineering problems
- Synthesize solutions to tasks in AI and critically evaluate alternatives using the knowledge and understanding of appropriate principles and guidelines
- Construct simple AI systems using a declarative programming language.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

	Course Learning Outcomes (CLOs)	**Blooms Level (Optional)**
CLO1	Explain the relationship between probability theory and logic by discussing probabilistic reasoning, with a focus on Bayes' Theorem	C6 , A3, A4 P2
CLO2	Apply search to game-playing and to solve Constraint Satisfaction Problems.	C3
CLO3	Apply inference and resolution in propositional and first-order predicate logic to prove theorems	C3

b) Mapping of CLO with PLO

No.	**(CLOs)**	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**	**PLO10**	**PLO11**	**PLO12**
1	CLO1	X											
2	CLO2			X									
3	CLO3			X									

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Introduction to Artificial Intelligence	• Lecture	Class Test Final Exam	CLO-1
Week -02	Intelligent agents	• Lecture	Class Test Final Exam	CLO-1
Week -03	Search techniques in AI	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2
Week -04	Uninformed, Informed/Heuristic search	• Lecture • Assignments	Class Test Final Exam	CLO-2
Week -05	Constraint satisfaction problems	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2
Week -06	Game Playing, Knowledge representation and reasoning	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-2
Week -07	Predicate Logic	• Lecture • Assignments	Class Test Final Exam	CLO-3

			Assignments	
Week -08	Planing	• Lecture	Class Test Final Exam	CLO-3
Week -09	Uncertainty and Probabilistic reasoning	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-3
Week -10	Expert system and Knowledge Engineering	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-1
Week -11	Learning	• Lecture • Assignments	Class Test Final Exam Assignments	CLO-1
Week -12	Natural Language Processing	• Lecture	Final Exam	CLO-1
Week -13	Computer Vision and Image Understanding	• Lecture	Final Exam	CLO-1

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Class participation and attendance	10%
Class tests / Class assessment	20%
Term Final Examination (3 hours duration)	70%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 30%
b) Summative: 70%
c) Make-up Procedures:
- Feedback on continuous assessment is given to the students immediately after the test.
- The minimum number of class-test/assignments are (n+1) with best n will be counted (here, n is number of credit). Based on the students' feedback additional class-test/assignment may be taken by the course teacher

Part D

16. Learning Materials

1) Recommended Readings

- Artificial Intelligence
  - Elaine Rich and Kevin Knight
- Artificial Intelligence A Modern Approach
  - Stuart Russell and Peter Norvig
- Introduction to Artificial Intelligence & Expert Systems
  - Dan W. Patterson

2) Others

- Handout/lecture material provided by the course teacher